Before Nomenclature

- Predicting ionic charges of elements
- Oxidation number: the electron number which an atom gives or takes when become compound

For example NaCl → Na^{+1} Cl^{-1}

$MgCl_2$ → Mg^{+2} Cl^{-1}

H_2O → H^{+1} O^{-2}

Rules for determining oxidation numbers

(If two rules contradict with each other, use above rule to determine.)

1- For free element atoms (did not form a compound with other atoms)
Oxidation number: 0
Cl ⟶ 0
Al ⟶ 0

2- a) For nötr species summation of oxidation number: 0
$[CH_3OH]$ ⟶ 0
$[MgCl_2]$ ⟶ 0

b) For an ion oxidation number is equal to charge of ion
Fe^{3+} ⟶ +3
MnO_4^- ⟶ -1

3- Group 1 metals have oxidation number +1

Group 2 metals have oxidation number +2 at their compounds

KCl $\longrightarrow$ K^{+1}

K_2CO_3 $\longrightarrow$ K^{+1}

$MgBr_2$ and $Mg(NO_3)_2$ $\longrightarrow$ Mg^{+2}

4- Fluorine at its compounds oxidation number is -1

HF $\longrightarrow$ F^{-1}

SF_6 $\longrightarrow$ F^{-1}

5- Hydrogen at its compounds oxidation number +1

CH_4 $\longrightarrow$ H^{+1}

NH_3 $\longrightarrow$ H^{+1}

H_2S $\longrightarrow$ H^{+1}

6- Oxygen at its compounds oxidation number -2

H_2O, CO_2

7- For binary compounds of metals

with group 7 elements → oxidation number -1

with group 6 elements → oxidation number -2

With group 5 elements → oxidation number -3

$MgBr_2 \longrightarrow Br^{-1}$

$Li_2S \longrightarrow S^{-2}$

$Li_3N \longrightarrow N^{-3}$

Find the oxidation number of underlined elements.

P_4 → Rule 1, P ox. number: 0

Al_2O_3 → Rule 2, summation of ox. number: 0
Rule 6, O ox. number: -2
Al ox. number: +3

MnO_4^- → Rule 2b, summation of ox. number of an ion is charge of the ion: -1
O ox. number: -2
Mn ox. number: +7

NaH → Rule 3, Na ox. Number: +1
Rule 5, H ox. Number: +1
if both of them +1 then $(NaH)^{2+}$! → This contradicts with Rule 2a
Rule 2 and Rule 3 is above Rule 5, so
Na ox. number: +1
H ox. number: -1

H_2O_2 → Rule 5, H ox. number: +1
O ox. number: -1
(Rule 5 is above Rule 6)

Exercises

- Below matters pure substance or mixture?

if mixture is it homogeneous or heterogeneous

A breath of fresh air
Orange juice
ice

- Below data is given for compounds formed between N and H

Compound	N (mass, g)	H (mass, g)
A	0.50	0.108
B	1.00	0.072
C	0.75	0.108

if B= N_2H_2 What is A and C?

Compound	N (mass, g)	H (mass, g)	
A	1.00	0.216/0.072 = 3	N_2H_6
B	1.00	0.072/0.072 = 1	N_2H_2
C	1.00	0.144/0.072 = 2	N_2H_4

- Determine the oxidation numbers for underlined elements

$\underline{C}H_4$, $\underline{S}F_4$, $\underline{Na}_2\underline{O}_2$, $\underline{Fe}O_4^{2-}$

- Name the given compunds

$Fe_2(SO_4)_3$

$Cu(OH)_2$

$KClO_4$

$Co(NO_3)_2$

HIO_3

SiF_4